

Terra Nova

DEVOPS TOOLS & CLOUD COMPUTING - Global Solution

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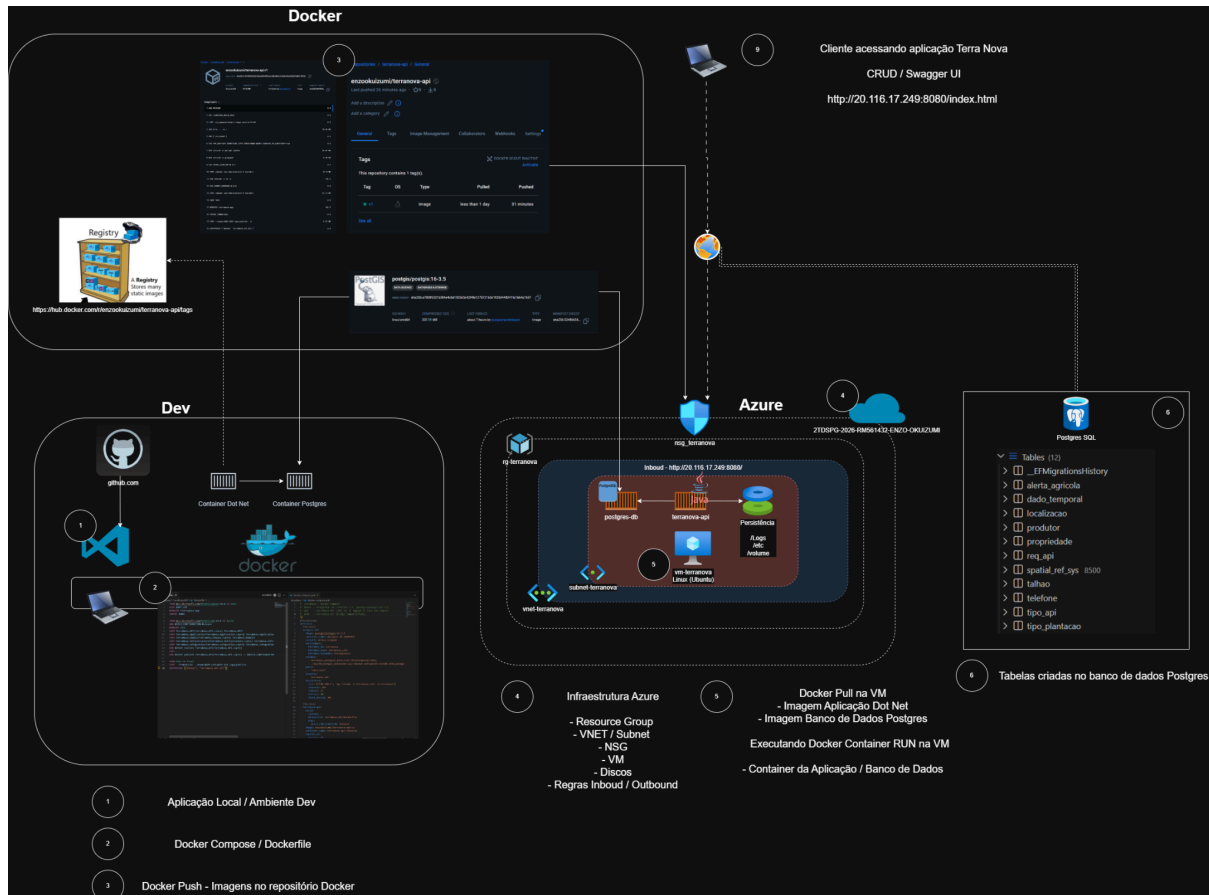
Link Vídeo Youtube + Repositório Git.....	3
Desenho da Arquitetura.....	3
Organização das Tarefas via Trello.....	4
Imagens DockerHub.....	5
Passo a passo na minha Aplicação no Azure CLI.....	6

Link Vídeo Youtube + Repositório Git

youtube: [Video Global Solution DevOps - Terra Nova](#)

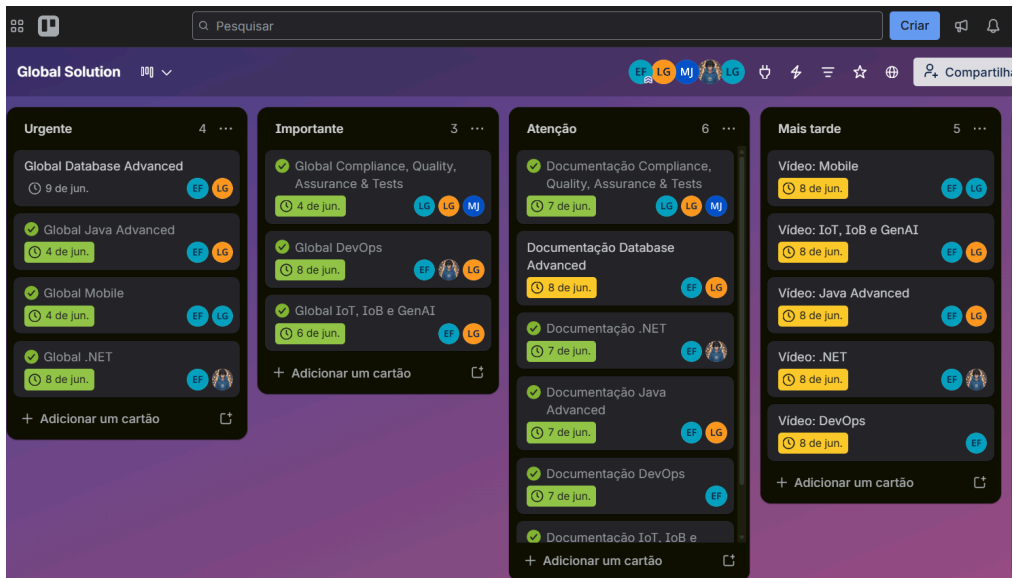
github: [Global-Solution-Space/DevOps-Tools-Cloud-Computing](#)

Desenho da Arquitetura



Link Diagrama: [gs-terranova-drawio](#)

Organização das Tarefas via Trello



Principal Responsável: Enzo Okuizumi, Gustavo Okada, Lucas Barros

Imagens DockerHub

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enzookuizumi/terranoa-api

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v1		Image	less than 1 day	31 minutes

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enzookuizumi/terranoa-api:v1

INDEX DIGEST sha256:1b9759255d3667a8aa55249f25aac36fa40e0cfa84e43a66668d7638319995e

OS/ARCH
linux/amd64

COMPRESSED SIZE
95.36 MB

LAST PUSHED
6 minutes by [enzookuizumi](#)

TYPE
Image

MANIFEST DIGEST
sha256:b8ef9b6b...

Image Layers

1 ARG RELEASE	0 B
2 ARG LAUNCHPAD_BUILD_ARCH	0 B
3 LABEL org.opencontainers.image.version=24.04	0 B
4 ADD file ... in /	28.36 MB
5 CMD ["/bin/bash"]	0 B
6 ENV APP_UID=1654 ASPNETCORE_HTTP_PORTS=8080 DOTNET_RUNNING_IN_CONTAINER=true	0 B
7 RUN /bin/sh -c apt-get update	16.04 MB
8 RUN /bin/sh -c groupadd	3.49 KB
9 ENV DOTNET_VERSION=10.0.8	0 B
10 COPY /dotnet /usr/share/dotnet # buildkit	34.9 MB
11 RUN /bin/sh -c ln -s	156 B
12 ENV ASPNET_VERSION=10.0.8	0 B
13 COPY /dotnet /usr/share/dotnet # buildkit	12.13 MB
14 USER 1654	0 B
15 WORKDIR /terranoa-app	106 B
16 EXPOSE [8080/tcp]	0 B
17 COPY --chown=1654:1654 /app/publish . #	3.93 MB
18 ENTRYPOINT ["dotnet" "TerraNova.API.dll"]	0 B

Passo a passo na minha Aplicação no Azure CLI

Logando no azure

```
Administrator: Windows PowerShell
fwlink/?linkid=2271136

Retrieving tenants and subscriptions for the selection...

[Tenant and subscription selection]

No      Subscription name      Subscription ID      Tenant
-----
[1]      2TDSFG-2026-RM561432-ENZO-OKUIZUMI  01bd2f0c-367c-4533-bf77-c3454ada4a18  Grupo Alun
[2]      2tdspg-rm564836-MiltonMarcelino      39ffaa05-1ee0-46d8-bb89-cdabdbba4288  Grupo Alun
[3] *    2tdspg-rm566422-LucasBarrosGouveia  213b06ed-e7f7-4f7b-a22d-f0f15daa85ff  Grupo Alun

The default is marked with an *; the default tenant is 'Grupo Alun' and subscription is '2tdspg-rm566422-LucasBarrosGouveia' (213b06ed-e7f7-4f7b-a22d-f0f15daa85ff).

Select a subscription and tenant (Type a number or Enter for no changes): 1

Tenant: Grupo Alun
Subscription: 2TDSFG-2026-RM561432-ENZO-OKUIZUMI (01bd2f0c-367c-4533-bf77-c3454ada4a18)

[Announcements]
With the new Azure CLI login experience, you can select the subscription you want to use more easily. Learn more about it and its configuration at https://go.microsoft.com/fwlink/?linkid=2271236

If you encounter any problem, please open an issue at https://aka.ms/azclibug

[Warning] The login output has been updated. Please be aware that it no longer displays the full list of available subscriptions by default.

PS C:\windows\system32>
```

Azure account show

```
Administrator: Windows PowerShell

Tenant: Grupo Alun
Subscription: 2TDSFG-2026-RM561432-ENZO-OKUIZUMI (01bd2f0c-367c-4533-bf77-c3454ada4a18)

[Announcements]
With the new Azure CLI login experience, you can select the subscription you want to use more easily. Learn more about it and its configuration at https://go.microsoft.com/fwlink/?linkid=2271236

If you encounter any problem, please open an issue at https://aka.ms/azclibug

[Warning] The login output has been updated. Please be aware that it no longer displays the full list of available subscriptions by default.

PS C:\windows\system32> az account show
{
  "environmentName": "AzureCloud",
  "homeTenantId": "11dbbfe2-89b8-4549-be10-cec364e59551",
  "id": "01bd2f0c-367c-4533-bf77-c3454ada4a18",
  "isDefault": true,
  "managedByTenants": [],
  "name": "2TDSFG-2026-RM561432-ENZO-OKUIZUMI",
  "state": "Enabled",
  "tenantDefaultDomain": "fiap.com.br",
  "tenantDisplayName": "Grupo Alun",
  "tenantId": "11dbbfe2-89b8-4549-be10-cec364e59551",
  "user": {
    "name": "RM561432@fiap.com.br",
    "type": "user"
  }
}

PS C:\windows\system32> cd
```

Rodando Azure CLI script

```
C:\Program Files\Microsoft SDKs\Azure\CLI2\python.exe
PS C:\Users\Enzo\new_backup\FIAP\_Projetos\Global Solution\DevOps-Tools-Cloud-Computing> bash azure-cli-script.
[1/6] Criando Grupo de Recursos...
{
  "id": "/subscriptions/01bd2f0c-367c-4533-bf77-c3454ada4a18/resourceGroups/rg-terranova",
  "location": "canadacentral",
  "managedBy": null,
  "name": "rg-terranova",
  "properties": {
    "provisioningState": "Succeeded"
  },
  "tags": null,
  "type": "Microsoft.Resources/resourceGroups"
}
[2/6] Criando VNet e Subnet...
```

Criando VNET e Subnet

```
C:\Program Files\Microsoft SDKs\Azure\CLI2\python.exe
{
  "sourcePortRange": "*",
  "sourcePortRanges": [],
  "type": "Microsoft.Network/networkSecurityGroups/securityRules"
},
{
  "access": "Allow",
  "destinationAddressPrefix": "*",
  "destinationAddressPrefixes": [],
  "destinationPortRange": "5432",
  "destinationPortRanges": [],
  "direction": "Inbound",
  "etag": "W/\"bf6115f9-e046-450c-9bd6-e5886075ace7\"",
  "id": "/subscriptions/01bd2f0c-367c-4533-bf77-c3454ada4a18/resourceGroups/rg-terranova/providers/Microsoft.Network/networkSecurityGroups/nsg-terranova/securityRules/open-port-5432",
  "name": "open-port-5432",
  "priority": 1020,
  "protocol": "*",
  "provisioningState": "Succeeded",
  "resourceGroup": "rg-terranova",
  "sourceAddressPrefix": "*",
  "sourceAddressPrefixes": [],
  "sourcePortRange": "*",
  "sourcePortRanges": [],
  "type": "Microsoft.Network/networkSecurityGroups/securityRules"
}
},
"type": "Microsoft.Network/networkSecurityGroups"
}
[6/6] Instalando Docker e dependencias na VM...
- Running ..
```

Docker e dependências

```
C:\Program Files\Microsoft SDKs\Azure\CLI2\python.exe
"type": "Microsoft.Network/networkSecurityGroups"
}
[6/6] Instalando Docker e dependências na VM...
{
  "value": [
    {
      "code": "ProvisioningState/succeeded",
      "displayStatus": "Provisioning succeeded",
      "level": "Info",
      "message": "Enable succeeded: \n[stdout]\namd64 pigz amd64 2.6-1 [63.6 kB]\nGet:2 https://download.docker.com/linux/ubuntu jammy/stable amd64 containerd.io amd64 2.2.4-1~ubuntu.22.04~jammy [23.6 MB]\nGet:3 https://download.docker.com/linux/ubuntu jammy/stable amd64 docker-ce-cli amd64 5:29.5.3-1~ubuntu.22.04~jammy [16.8 MB]\nGet:4 https://download.docker.com/linux/ubuntu jammy/stable amd64 docker-ce amd64 5:29.5.3-1~ubuntu.22.04~jammy [23.0 MB]\nGet:5 https://download.docker.com/linux/ubuntu jammy/stable amd64 docker-buildx-plugin amd64 0.34.1-1~ubuntu.22.04~jammy [17.2 MB]\nGet:6 https://download.docker.com/linux/ubuntu jammy/stable amd64 docker-ce-rootless-extras amd64 5:29.5.3-1~ubuntu.22.04~jammy [10.2 MB]\nGet:7 https://download.docker.com/linux/ubuntu jammy/stable amd64 docker-compose-plugin amd64 5.1.4-1~ubuntu.22.04~jammy [8126 kB]\nFetched 98.9 MB in 1s (159 MB/s)\nSelecting previously unselected package containerd.io.\n(Reading database ... \n(Reading database ... 5%\n(Reading database ... 10%\n(Reading database ... 15%\n(Reading database ... 20%\n(Reading database ... 25%\n(Reading database ... 30%\n(Reading database ... 35%\n(Reading database ... 40%\n(Reading database ... 45%\n(Reading database ... 50%\n(Reading database ... 55%\n(Reading database ... 60%\n(Reading database ... 65%\n(Reading database ... 70%\n(Reading database ... 75%\n(Reading database ... 80%\n(Reading database ... 85%\n(Reading database ... 90%\n(Reading database ... 95%\n(Reading database ... 100%\n(Reading database ... 62915 files and directories currently installed.)\nPreparing to unpack .../0-containerd.io_2.2.4-1~ubuntu.22.04~jammy_amd64.deb ...\nUnpacking containerd.io (2.2.4-1~ubuntu.22.04~jammy) ...\nSelecting previously unselected package docker-ce-cli.\nPreparing to unpack .../1-docker-ce-cli_5%3a29.5.3-1~ubuntu.22.04~jammy_amd64.deb ...\nUnpacking docker-ce-cli (5:29.5.3-1~ubuntu.22.04~jammy) ...\nSelecting previously unselected package docker-ce.\nPreparing to unpack .../2-docker-ce_5%3a29.5.3-1~ubuntu.22.04~jammy_amd64.deb ...\nUnpacking docker-ce (5:29.5.3-1~ubuntu.22.04~jammy) ...\nSelecting previously unselected package pigz.\nPreparing to unpack .../3-pigz_2.6-1_amd64.deb ...\nUnpacking pigz (2.6-1) ...\nSelecting previously unselected package docker-buildx-plugin.\nPreparing to unpack .../4-docker-buildx-plugin_0.34.1-1~ubuntu.22.04~jammy_amd64.deb ...\nUnpacking docker-buildx-plugin (0.34.1-1~ubuntu.22.04~jammy) ...\nSelecting previously
```

Criando VM Linux

```
[4/6] Criando Máquina Virtual Linux...
The default value of '--size' will be changed to 'Standard_D2s_v5' from 'Standard_DS1_v2' in a future release.
Use existing SSH public key file: C:\Users\Enzo\.ssh\id_rsa.pub
Consider upgrading security for your workloads using Azure Trusted Launch VMs. To know more about Trusted Launch, please visit https://aka.ms/TrustedLaunch.
{
  "fqdns": "",
  "id": "/subscriptions/01bd2f0c-367c-4533-bf77-c3454ada4a18/resourceGroups/rg-challenge-petguardian/providers/Microsoft.Compute/virtualMachines/vm-petguardian",
  "location": "canadacentral",
  "macAddress": "00-0D-3A-FF-12-52",
  "powerState": "VM running",
  "privateIpAddress": "10.10.1.4",
  "publicIpAddress": "20.104.24.205",
  "resourceGroup": "rg-challenge-petguardian"
}
Command ran in 46.058 seconds (init: 0.223, invoke: 45.834)
```


Liberando Portas Necessárias (8080 e 5432)

```
C:\Program Files\Microsoft SDKs\Azure\CLI2\python.exe
{
  "etag": "W/\"abbab288-8098-44e3-a165-f0febb24743b\"\"",
  "id": "/subscriptions/01bd2f0c-367c-4533-bf77-c3454ada4a18/resourceGroups/rg-terranova/providers/Microsoft.Network/virtualNetworks/vnet-terranova",
  "location": "canadacentral",
  "name": "vnet-terranova",
  "privateEndpointVNetPolicies": "Disabled",
  "provisioningState": "Succeeded",
  "resourceGroup": "rg-terranova",
  "resourceGuid": "7ec27627-54af-4ae5-9ecd-22f3fa5ca621",
  "subnets": [
    {
      "addressPrefix": "10.20.1.0/24",
      "delegations": [],
      "etag": "W/\"abbab288-8098-44e3-a165-f0febb24743b\"\"",
      "id": "/subscriptions/01bd2f0c-367c-4533-bf77-c3454ada4a18/resourceGroups/rg-terranova/providers/Microsoft.Network/virtualNetworks/vnet-terranova/subnets/subnet-terranova",
      "name": "subnet-terranova",
      "privateEndpointNetworkPolicies": "Disabled",
      "privateLinkServiceNetworkPolicies": "Enabled",
      "provisioningState": "Succeeded",
      "resourceGroup": "rg-terranova",
      "type": "Microsoft.Network/virtualNetworks/subnets"
    }
  ],
  "type": "Microsoft.Network/virtualNetworks",
  "virtualNetworkPeerings": []
}
3/6] Criando Network Security Group...
```

Instalando Docker e Dependências

```
Administrator: Windows PowerShell
6/6] Instalando Docker e Dependências...

"value": [
  {
    "code": "ProvisioningState/succeeded",
    "displayStatus": "Provisioning succeeded",
    "level": "Info",
    "message": "Enable succeeded: \n[stdout]\n78 MB of additional disk space will be used.\nGet:1 http://azure.ubuntu.com/ubuntu jammy/universe amd64 pigz amd64 2.6-1 [63.6 kB]\nGet:2 https://download.docker.com/linux/ubuntu jammy/stable amd64 containerd.io amd64 2.2.4-1~ubuntu.22.04~jammy [23.6 MB]\nGet:3 https://download.docker.com/linux/ubuntu jammy/stable amd64 docker-ce-cli amd64 5:29.5.2-1~ubuntu.22.04~jammy [17.0 MB]\nGet:4 https://download.docker.com/linux/ubuntu jammy/stable amd64 docker-ce amd64 5:29.5.2-1~ubuntu.22.04~jammy [23.0 MB]\nGet:5 https://download.docker.com/linux/ubuntu jammy/stable amd64 docker-buildx-plugin amd64 0.34.0-1~ubuntu.22.04~jammy [17.2 MB]\nGet:6 https://download.docker.com/linux/ubuntu jammy/stable amd64 docker-ce-rootless-extras amd64 5:29.5.2-1~ubuntu.22.04~jammy [10.2 MB]\nGet:7 https://download.docker.com/linux/ubuntu jammy/stable amd64 docker-compose-plugin amd64 5.1.4-1~ubuntu.22.04~jammy [13.8 MB]\nFetched 99.1 MB in 1s (134 MB/s)\nSelecting previously unselected package containerd.io.\n(Reading database ... 5%\r(Reading database ... 10%\r(Reading database ... 15%\r(Reading database ... 20%\r(Reading database ... 25%\r(Reading database ... 30%\r(Reading database ... 35%\r(Reading database ... 40%\r(Reading database ... 45%\r(Reading database ... 50%\r(Reading database ... 55%\r(Reading database ... 60%\r(Reading database ... 65%\r(Reading database ... 70%\r(Reading database ... 75%\r(Reading database ... 80%\r(Reading database ... 85%\r(Reading database ... 90%\r(Reading database ... 95%\r(Reading database ... 100%\r(Reading database ... 62898 files and directories currently installed.)\nPreparing to unpack .../0-containerd.io_2.2.4-1~ubuntu.22.04~jammy_amd64.deb ...\nUnpacking containerd.io (2.2.4-1~ubuntu.22.04~jammy) ...\nSelecting previously unselected package docker-ce-cli.\nPreparing to unpack .../1-docker-ce-cli_5%3a29.5.2-1~ubuntu.22.04~jammy_amd64.deb ...\nUnpacking docker-ce-cli (5:29.5.2-1~ubuntu.22.04~jammy) ...\nSelecting previously unselected package docker-ce.\nPreparing to unpack .../2-docker-ce_5%3a29.5.2-1~ubuntu.22.04~jammy_amd64.deb ...\nUnpacking docker-ce (5:29.5.2-1~ubuntu.22.04~jammy) ...\nSelecting previously unselected package pigz.\nPreparing to unpack .../3-pigz_2.6-1_amd64.deb ...\nUnpacking pigz (2.6-1) ...\nSelecting previously unselected package docker-buildx-plugin.\nPreparing to unpack .../4-docker-buildx-plugin_0.34.0-1~ubuntu.22.04~jammy_amd64.deb ...\nUnpacking docker-buildx-plugin (0.34.0-1~ubuntu.22.04~jammy) ...\nSelecting previously unselected package docker-compose-plugin.\nPreparing to unpack .../5-docker-compose-plugin_5.1.4-1~ubuntu.22.04~jammy_amd64.deb ...\nUnpacking docker-compose-plugin (5.1.4-1~ubuntu.22.04~jammy) ...
```

Provisionamento da VM Concluído

```
ntrolling tty.)\ndebug: falling back to frontend: teletype\ndpkg-preconfigure: unable to re-open stdin:
ble to initialize frontend: Dialog\ndebug: (Dialog frontend will not work on a dumb terminal, an emacs
r without a controlling terminal.)\ndebug: falling back to frontend: Readline\ndebug: unable to ini
Readline\ndebug: (This frontend requires a controlling tty.)\ndebug: falling back to frontend: Tel
nfigure: unable to re-open stdin: \nSynchronizing state of docker.service with SysV service script with
temd-sysv-install.\nExecuting: /lib/systemd/systemd-sysv-install enable docker\n"
}
]
}
=====
Provisionamento concluido com sucesso!
IP Publico da VM : 20.116.17.249
Acesse via SSH : ssh terranova-admin@20.116.17.249
:8080 : http://20.116.17.249
:8080/index.html : http://20.116.17.249
:5432 (db: terranova / user: terranova_user)
=====
PS C:\Users\Enzo\new_backup\FIAP\Projetos\Global Solution\DevOps-Tools-Cloud-Computing>
```

VM no Azure

Name ↑	Subscription	Location
 NetworkWatcherRG	... 2tdspg-rm566422-LucasBarrosGouveia	South Africa North
 NetworkWatcherRG	... 2tdspg-rm564836-MiltonMarcelino	Canada Central
 rg-terranova	... 2TDSFG-2026-RM561432-ENZO-OKUIZUMI	Canada Central
 rg_Challenge_Wise_Clyvo_VET	... 2tdspg-rm564836-MiltonMarcelino	Brazil South
 rg_CP3_DevOps	... 2tdspg-rm566422-LucasBarrosGouveia	South Africa North
 vm-devops-cp3_group	... 2tdspg-rm566422-LucasBarrosGouveia	South Africa North
 vm-devops-linux-rm561432_group	... 2TDSFG-2026-RM561432-ENZO-OKUIZUMI	Canada Central
 vm-devops_group	... 2tdspg-rm564836-MiltonMarcelino	Canada Central

Resource Group

☐	Name ↑		Type	Location
☐	 nsg-terranova	...	Network securit...	Canada Central
☐	 vm-terranova	...	Virtual machine	Canada Central
☐	 vm-terranova_OsDisk_1_3d01a380		Disk	Canada Central
☐	 vm-terranovaPublicIP	...	Public IP address	Canada Central
☐	 vm-terranovaVMNic	...	Network Interface	Canada Central
☐	 vnet-terranova	...	Virtual network	Canada Central

Entrando na VM

```
=====
PS C:\Users\Enzo\new_backup\FIAP\Projetos\Global Solution\DevOps-Tools-Cloud-Computing> ssh terranova-adm@20.116.17.249
The authenticity of host '20.116.17.249 (20.116.17.249)' can't be established.
ED25519 key fingerprint is SHA256:Va5s5u48gcYMFGv+9mLrPQf4r9P36Vrf2T4l3vZyHuM.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '20.116.17.249' (ED25519) to the list of known hosts.
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1052-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Jun  8 19:32:11 UTC 2026

System load:  0.45          Processes:           139
Usage of /:   8.1% of 28.89GB Users logged in:       0
Memory usage: 9%           IPv4 address for eth0: 10.20.1.4
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

51 updates can be applied immediately.
31 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

terranova-adm@vm-terranova:~$
```

Clonando Repositório

```
terranova-adm@vm-terranova:~$ git clone https://github.com/Global-Solution-Space/DevOps-Tools-Cloud-Computing
Cloning into 'DevOps-Tools-Cloud-Computing'...
remote: Enumerating objects: 527, done.
remote: Counting objects: 100% (527/527), done.
remote: Compressing objects: 100% (302/302), done.
remote: Total 527 (delta 297), reused 437 (delta 210), pack-reused 0 (from 0)
Receiving objects: 100% (527/527), 127.75 KiB | 3.76 MiB/s, done.
Resolving deltas: 100% (297/297), done.
terranova-adm@vm-terranova:~$ ls
DevOps-Tools-Cloud-Computing
terranova-adm@vm-terranova:~$ ls -la
total 32
drwxr-x--- 5 terranova-adm terranova-adm 4096 Jun  8 19:41 .
drwxr-xr-x 3 root          root          4096 Jun  8 19:29 ..
-rw-r--r-- 1 terranova-adm terranova-adm 220 Jan  6 2022 .bash_logout
-rw-r--r-- 1 terranova-adm terranova-adm 3771 Jan  6 2022 .bashrc
drwx----- 2 terranova-adm terranova-adm 4096 Jun  8 19:32 .cache
-rw-r--r-- 1 terranova-adm terranova-adm 807 Jan  6 2022 .profile
drwx----- 2 terranova-adm terranova-adm 4096 Jun  8 19:29 .ssh
drwxrwxr-x 4 terranova-adm terranova-adm 4096 Jun  8 19:41 DevOps-Tools-Cloud-Computing
terranova-adm@vm-terranova:~$
```

Rodando Docker Compose up -d

```
terranova-adm@vm-terranova:~/DevOps-Tools-Cloud-Computing$ cd TerraNova/
terranova-adm@vm-terranova:~/DevOps-Tools-Cloud-Computing/TerraNova$ docker compose up -d --build
[+] up 20/20
[+] Image postgres/postgis:16-3.5 Pulled 8.6s
[+] Building 1.4s (6/19)
=> [internal] load local bake definitions 0.0s
=> => reading from stdin 686B 0.0s
=> [internal] load build definition from Dockerfile 0.1s
=> => transferring dockerfile: 898B 0.0s
=> [internal] load metadata for mcr.microsoft.com/dotnet/sdk:10.0 0.2s
=> [internal] load metadata for mcr.microsoft.com/dotnet/aspnet:10.0 0.3s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [build 1/10] FROM mcr.microsoft.com/dotnet/sdk:10.0@sha256:c0790639332692a0d56cdd81ed581cfd24d040d9839764c13 0.9s
=> => resolve mcr.microsoft.com/dotnet/sdk:10.0@sha256:c0790639332692a0d56cdd81ed581cfd24d040d9839764c138994866d 0.1s
=> [base 1/2] FROM mcr.microsoft.com/dotnet/aspnet:10.0@sha256:8c0b6857eab7b2aa57884c839bf4678414606bd7d17370f18 0.9s
=> => resolve mcr.microsoft.com/dotnet/aspnet:10.0@sha256:8c0b6857eab7b2aa57884c839bf4678414606bd7d17370f18a842a 0.1s
=> => sha256:a31473e9874a809f6ebc22a88375800bc9bad0e8cef7bf18c94d0c4bc2274954 156B / 156B 0.1s
=> => sha256:b6cfff015c25a7d5360008f7f5f32d556977ed1679adebcff189c6e8dda692f1 3.57kB / 3.57kB 0.1s
=> => sha256:2d0efc714cbde17ddf1dc365e270070659acacc9eb813fe10222ffdec3a8a90 12.72MB / 12.72MB 0.6s
=> => sha256:ffe6f39219bf9e567d7272187711e20261019149c6af5785203b668c2e2351ba 34.60MB / 36.59MB 0.5s
=> => sha256:58abd912a5fb736d822930426374c79bbf5a65927dac845192aec63ee6fe1b9e 16.82MB / 16.82MB 0.5s
=> => sha256:cb259a83ac3dd9fea0b394df41df2b298adf0df938fef5999475af18a751c257 19.92MB / 29.73MB 0.4s
=> [internal] load build context 0.1s
=> => transferring context: 246.73kB 0.0s
```

Criação / Execução dos Container


```
terranoa-adm@vm-terranoa: ~/DevOps-Tools-Cloud-Computing/TerraNova
-> => extracting sha256:2d0efc714cbde17ddf1dc365e270070659acacc9eb813fe10222ffdec3a8a90
-> [internal] load build context
-> => transferring context: 246.73kB
-> [base 2/2] WORKDIR /terranoa-app
-> [build 2/10] WORKDIR /src
-> [build 3/10] COPY TerraNova.API/TerraNova.API.csproj TerraNova.API/ 0.1s
-> [build 4/10] COPY TerraNova.Application/TerraNova.Application.csproj TerraNova.Application/ 0.1s
-> [build 5/10] COPY TerraNova.Domain/TerraNova.Domain.csproj TerraNova.Domain/ 0.1s
-> [build 6/10] COPY TerraNova.Infrastructure/TerraNova.Infrastructure.csproj TerraNova.Infrastructure/ 0.1s
-> [build 7/10] COPY TerraNova.Integration/TerraNova.Integration.csproj TerraNova.Integration/ 0.1s
-> [build 8/10] RUN dotnet restore TerraNova.API/TerraNova.API.csproj 6.8s
-> [build 9/10] COPY . . 0.1s
-> [build 10/10] RUN dotnet publish TerraNova.API/TerraNova.API.csproj -c Release -o /app/publish /p:UseAppHost=false 14.9s
-> [final 1/1] COPY --from=build --chown=1654:1654 /app/publish . 0.1s
-> exporting to image
-> => exporting layers 0.6s
-> => exporting manifest sha256:0fe34c5aaa211dd33cd6a7e8957dffa6c6db112bbc3ef2bbaf6680fc812456 0.0s
-> => exporting config sha256:4af03329af96a4ab6b9b9430677dbf4cd4720c202936b8e7b3f0698f6bb0802c 0.0s
-> => exporting attestation manifest sha256:31b1f2e6a79c241f1468c489f758a3c474ca290789c77b5048ef94e65a297199 0.0s
-> => exporting manifest list sha256:b0b67d3874a43c5bf945071ff09ffe856e9e6169613843a967c53a18fa26a007 0.0s
-> => naming to docker.io/enzookuizumi/terranoa-api:v1 0.0s
[+] up 25/25king to docker.io/enzookuizumi/terranoa-api:v1 0.1s
✔Image postgres/postgis:16-3.5 Pulled 8.6s
✔Image enzookuizumi/terranoa-api:v1 Built 40.7s
✔Network terranoa_terranoa_net Created 0.0s
✔Volume terranoa_terranoa_postgres_data Created 0.0s
✔Container postgres-db-rm561432 Healthy 6.0s
✔Container terranoa-api-rm561432 Started 6.0s
terranoa-adm@vm-terranoa:~/DevOps-Tools-Cloud-Computing/TerraNova$ ^[[200~docker compose ps~
docker: command not found
terranoa-adm@vm-terranoa:~/DevOps-Tools-Cloud-Computing/TerraNova$ docker compose ps
NAME                IMAGE                                COMMAND                                SERVICE        CREATED        STATUS
postgres-db-rm561432 postgres/postgis:16-3.5              "docker-entrypoint.s..." postgres-db     36 seconds ago Up 35 seconds (healthy)
terranoa-api-rm561432 enzookuizumi/terranoa-api:v1         "dotnet TerraNova.AP..." terranoa-api   36 seconds ago Up 29 seconds
```

Docker compose logs -f / whoami / pwd

```
app@0b6ce1d97480: /terrano-app
postgres-db-rm561432 | CREATE EXTENSION
postgres-db-rm561432 | You are now connected to database "terrano" as user "terrano_user".
postgres-db-rm561432 | DO
postgres-db-rm561432 |
postgres-db-rm561432 | waiting for server to shut down...2026-06-08 19:53:09.553 UTC [47] LOG: received fast shutdown request
postgres-db-rm561432 | 2026-06-08 19:53:09.557 UTC [47] LOG: aborting any active transactions
postgres-db-rm561432 | 2026-06-08 19:53:09.558 UTC [47] LOG: background worker "logical replication launcher" (PID 53) exited with exit code 1
postgres-db-rm561432 |
postgres-db-rm561432 | 2026-06-08 19:53:09.559 UTC [48] LOG: shutting down
postgres-db-rm561432 | 2026-06-08 19:53:09.562 UTC [48] LOG: checkpoint starting: shutdown immediate
postgres-db-rm561432 | 2026-06-08 19:53:10.184 UTC [48] LOG: checkpoint complete: wrote 4459 buffers (27.2%); 0 WAL file(s) added, 0 removed, 2 recycled; write=0.215 s, sync=0.391 s, total=0.625 s; sync files=963, longest=0.207 s, average=0.001 s; distance=34802 kB, estimate=34802 kB; lsn=0/36E3468, redo lsn=0/36E3468
postgres-db-rm561432 | 2026-06-08 19:53:10.190 UTC [47] LOG: database system is shut down
postgres-db-rm561432 | done
postgres-db-rm561432 | server stopped
postgres-db-rm561432 |
postgres-db-rm561432 | PostgreSQL init process complete; ready for start up.
postgres-db-rm561432 |
postgres-db-rm561432 | 2026-06-08 19:53:10.277 UTC [1] LOG: starting PostgreSQL 16.9 (Debian 16.9-1.pgdg110+1) on x86_64-pc-linux-gnu, compiled by gcc (Debian 10.2.1-6) 10.2.1 202110110, 64-bit
postgres-db-rm561432 | 2026-06-08 19:53:10.277 UTC [1] LOG: listening on IPv4 address "0.0.0.0", port 5432
postgres-db-rm561432 | 2026-06-08 19:53:10.278 UTC [1] LOG: listening on IPv6 address ":::", port 5432
postgres-db-rm561432 | 2026-06-08 19:53:10.292 UTC [1] LOG: listening on Unix socket "/var/run/postgresql/.s.PGSQL.5432"
postgres-db-rm561432 | 2026-06-08 19:53:10.306 UTC [73] LOG: database system was shut down at 2026-06-08 19:53:10 UTC
postgres-db-rm561432 | 2026-06-08 19:53:10.317 UTC [1] LOG: database system is ready to accept connections
postgres-db-rm561432 | 2026-06-08 19:53:12.705 UTC [86] ERROR: relation "__EFMigrationsHistory" does not exist at character 45
postgres-db-rm561432 | 2026-06-08 19:53:12.705 UTC [86] STATEMENT: SELECT "MigrationId", "ProductVersion"
postgres-db-rm561432 | FROM "__EFMigrationsHistory"
postgres-db-rm561432 | ORDER BY "MigrationId"
^C
terrano-admin@vm-terrano:~/DevOps-Tools-Cloud-Computing/TerraNova$ docker container exec -it terrano-api-rm561432 /bin/bash
app@0b6ce1d97480: /terrano-app$ whoami
app
app@0b6ce1d97480: /terrano-app$ pwd
/terrano-app
app@0b6ce1d97480: /terrano-app$
```

CRUD produtor na Aplicação Dot Net

```
terrano-admin@vm-terrano:~$ # CREATE (cadastra produtor + telefone na mesma chamada)
PRODUTOR_ID=$(curl -fsS -X POST "$API_URL/api/producao" \
-H "Content-Type: application/json" \
-d '{
  "nome": "João da Silva",
  "email": "joao.silva@terrano.com",
  "senha": "senha123",
  "telefoneContato": "11987654321"
}' | jq -r '.id // .Id')

echo "PRODUTOR_ID=$PRODUTOR_ID"

# READ ALL
curl -fsS "$API_URL/api/producao"

# READ BY ID
curl -fsS "$API_URL/api/producao/$PRODUTOR_ID"

# READ BY EMAIL
curl -fsS "$API_URL/api/producao/by-email?email=joao.silva@terrano.com"

# UPDATE
curl -fsS -X PUT "$API_URL/api/producao/$PRODUTOR_ID" \
-H "Content-Type: application/json" \
-d '{
  "nome": "João da Silva Jr.",
  "email": "joao.jr@terrano.com",
  "senha": "novaSenha123",
  "telefoneContato": "11999998888"
}'

PRODUTOR_ID=b8ce1289-7207-400b-b0bc-86a8e2280552
[{"id":"a6c0cdae-0343-4863-act6-fc990fa675b","nome":"João da Silva Jr.", "email":"joao.jr1780950553@tn.com", "telefoneContato":"(11) 980950553"}, {"id":"b8ce1289-7207-400b-b0bc-86a8e2280552", "nome":"João da Silva", "email":"joao.silva@terrano.com", "telefoneContato":"(11) 987654321"}] [{"id":"b8ce1289-7207-400b-b0bc-86a8e2280552", "nome":"João da Silva", "email":"joao.silva@terrano.com", "telefoneContato":"(11) 987654321"}] [{"id":"b8ce1289-7207-400b-b0bc-86a8e2280552", "nome":"João da Silva", "email":"joao.silva@terrano.com", "telefoneContato":"(11) 987654321"}] [{"id":"b8ce1289-7207-400b-b0bc-86a8e2280552", "nome":"João da Silva Jr.", "email":"joao.jr@terrano.com", "telefoneContato":"(11) 999998888"}] terrano-admin@vm-terrano:~$
```

CRUD tipo_plantação na Aplicação Dot Net

```
terranova-adm@vm-terranova:~$ API_URL="http://20.116.17.249:8080"
terranova-adm@vm-terranova:~$
terranova-adm@vm-terranova:~$ # CREATE
TIPO_PLANTACAO_ID=$(curl -fsS -X POST "$API_URL/api/tipoplantacao" \
-H "Content-Type: application/json" \
-d '{
  "tipoPlant": "Soja"
}' | jq -r '.id // .Id')

echo "TIPO_PLANTACAO_ID=$TIPO_PLANTACAO_ID"

# READ ALL
curl -fsS "$API_URL/api/tipoplantacao"

# READ BY ID
curl -fsS "$API_URL/api/tipoplantacao/$TIPO_PLANTACAO_ID"

# UPDATE
curl -fsS -X PUT "$API_URL/api/tipoplantacao/$TIPO_PLANTACAO_ID" \
-H "Content-Type: application/json" \
-d '{
  "tipoPlant": "Soja Transgênica"
}'
TIPO_PLANTACAO_ID=f12c062b-0158-48b8-b1e7-6cf942c8bdf5
[{"id":"f12c062b-0158-48b8-b1e7-6cf942c8bdf5","tipoPlant":"Soja"}][{"id":"f12c062b-0158-48b8-b1e7-6# CREATEDf5","tipoPlant":"Soja Transgênica"}]terranova-adm@vm-terranova:~$ # CREATE
```

CRUD localizacao na Aplicação Dot Net

```
id":"f12c062b-0158-48b8-b1e7-6# CREATEDf5","tipoPlant":"Soja Transgênica"}]terranova-adm@vm-terranova:~$ # CREATE
LOCALIZACAO_ID=$(curl -fsS -X POST "$API_URL/api/localizacao" \
-H "Content-Type: application/json" \
-d '{
  "latitude": -23.5505,
  "longitude": -46.6333
}' | jq -r '.id // .Id')

echo "LOCALIZACAO_ID=$LOCALIZACAO_ID"

# READ ALL
curl -fsS "$API_URL/api/localizacao"

# READ BY ID
curl -fsS "$API_URL/api/localizacao/$LOCALIZACAO_ID"

# UPDATE
curl -fsS -X PUT "$API_URL/api/localizacao/$LOCALIZACAO_ID" \
-H "Content-Type: application/json" \
-d '{
  "latitude": -22.9068,
  "longitude": -43.1729
}'
LOCALIZACAO_ID=1542d2c9-348d-4139-83e4-bb4ed6b23c9e
[{"id":"079a8bc9-fc44-47d9-8908-5161bcd53693","latitude":-22.9068,"longitude":-43.1729},{"id":"1542d2c9-348d-4139-83e4-bb4ed6b23c9e","latitude":-23.5505,"longitude":-46.6333}][{"id":"1542d2c9-348d-4139-83e4-bb4ed6b23c9e","latitude":-23.5505,"longitude":-46.6333}][{"id":"1542d2c9-348d-4139-83e4-bb4ed6b23c9e","latitude":-22.9068,"longitude":-43.1729}]terranova-adm@vm-terranova:~$ # CREATE (cadastra
```

Verificando Dados persistidos no banco Oracle

```
terranoa-adm@vm-terranoa:~$ # DELETE PRODUTOR
curl -fsS -X DELETE "$API_URL/api/produtor/$PRODUTOR_ID"

# DELETE LOCALIZAÇÃO
curl -fsS -X DELETE "$API_URL/api/localizacao/$LOCALIZACAO_ID"

# DELETE TIPO DE PLANTACÃO
curl -fsS -X DELETE "$API_URL/api/tipoplantacao/$TIPO_PLANTACAO_ID"
terranoa-adm@vm-terranoa:~$ docker exec -it postgres-db-rm561432 psql -U terranoa_user -d terranoa
psql (16.9 (Debian 16.9-1.pgdg110+1))
Type "help" for help.

terranoa=# SELECT * FROM produtor;
      id_produto |      nome      |      email      |      senha
-----+-----+-----+-----
 a6c0cdae-0343-4863-ac1e-6fc990fa675b | João da Silva Jr. | joao.jr1780950553@tn.com | novaSenha123
(1 row)

terranoa=# SELECT * FROM localizacao;
      id_localizacao |      coordenadas
-----+-----
 079a8bc9-fc44-47d9-8908-5161bcd53693 | 0101000020E61000003CBD5296219645C04703780B24E836C0
(1 row)

terranoa=# select * FROM tipo_platancao;
terranoa=# select * FROM tipo_platancao;
ERROR:  syntax error at or near "select"
LINE 2: select * FROM tipo_platancao;
        ^

terranoa=# select * FROM tipo_platancao;
ERROR:  relation "tipo_platancao" does not exist
LINE 1: select * FROM tipo_platancao;
        ^

terranoa=# \dt
terranoa=# SELECT * FROM tipo_plantacao;
      id_tipo_plant | tipo_plant
-----+-----
(0 rows)

terranoa=#
```


Acessando Swagger da Aplicação Dot Net via IP da VM

The screenshot shows a web browser window with the Swagger UI for the TerraNova API v1. The browser's address bar shows the URL `20.116.17.249:8080/index.html`. The Swagger UI header includes the Swagger logo, the text "Select a definition", and a dropdown menu showing "TerraNova API v1".

The main content area displays the API title "TerraNova API" with a "v1" tag and a "OAS 3.0" badge. Below the title, it says "API REST para monitoramento agrícola com dados SatVeg e NASA POWER." and provides a link to "Contact Equipe TerraNova".

The API is organized into two sections:

- Produtor** (Gerenciamento de produtores rurais):
 - GET** `/api/Produtor`: Lista todos os produtores.
 - POST** `/api/Produtor`: Cadastra um novo produtor.
 - GET** `/api/Produtor/{id}`: Obtém um produtor pelo ID.
 - PUT** `/api/Produtor/{id}`: Atualiza um produtor pelo ID.
 - DELETE** `/api/Produtor/{id}`: Remove um produtor pelo ID.
 - GET** `/api/Produtor/by-email`: Obtém um produtor pelo e-mail.
- Telefone** (Gerenciamento de contatos telefônicos):
 - GET** `/api/Telefone`: Lista todos os registros de telefone cadastrados.